

ENTERPRISE INFRASTRUCTURE PLANNING FOR A STARTUP COMPANY

Week 2 Portfolio Project

ITEP 414 – System Administration and Maintenance
Bachelor of Science in Information Technology

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PART 1 – COMPANY PROFILE

Company Name

ABC Startup Solutions

Nature of Business

ABC Startup Solutions is a newly established software development company that provides customized software solutions for small and medium-sized businesses. The company develops web applications, business management systems, and other technology-based solutions that help organizations improve their daily operations.

Company Vision

To become a trusted software development company that provides reliable, innovative, and practical technology solutions to businesses while maintaining excellent service and professional standards.

Company Mission

ABC Startup Solutions aims to develop quality software solutions that address the needs of its clients. The company is committed to using appropriate technology, protecting business information, supporting employees, and continuously improving its services.

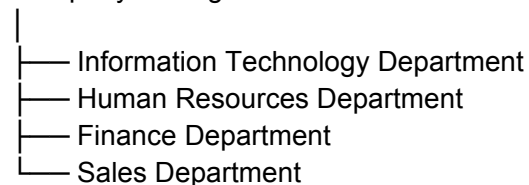
Office Location

2nd Floor, ABC Business Center, National Highway, Calamba City, Laguna, Philippines

The company occupies one office floor where all four departments operate. Since the company is newly established, its entire IT infrastructure will be planned and deployed from the beginning.

Organizational Structure

Company Manager / CEO



Employee Distribution

Department	Employees	Main Function
Information Technology	5	Software development, technical

		support, systems and network management
Human Resources	4	Recruitment, employee records, and administrative tasks
Finance	5	Accounting, budgeting, payroll, and financial records
Sales	6	Marketing, client communication, and sales activities
TOTAL	20	

PART 2 – ENTERPRISE HARDWARE INVENTORY

The hardware inventory is designed for 20 employees working in one office floor. The selection prioritizes reliable workstations, centralized services, network connectivity, power protection, and data backup.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	15	IT, HR, Finance, Sales	Primary workstation for employees
HW-002	Laptop	5	IT	Development, troubleshooting, and mobility
HW-003	Server	1	IT	Centralized file, application, and network services
HW-004	Router	1	IT	Manages network traffic and internet connectivity
HW-005	Managed Switch	1	IT	Connects wired computers and network devices
HW-006	Printer	2	HR, Finance	Printing documents, reports, and records
HW-007	UPS	4	IT / Server Room	Power protection for server and critical network equipment
HW-008	Wireless Access Point	3	Entire Office	Provides wireless network connectivity
HW-009	NAS Storage	1	IT	Centralized storage and backup
HW-010	External Backup Drive	2	IT	Offline backup of important company data
HW-011	Monitor	20	All Departments	Display for employee workstations

Hardware Selection Justification

The quantities are based on the company's 20 employees and four departments. Desktop computers serve as the primary workstations, while five laptops are assigned to IT employees for development, maintenance, and troubleshooting. A dedicated server and NAS provide centralized services and storage. The router, managed switch, and wireless access points provide network connectivity throughout the office. Printers support document-heavy departments, while UPS units and external backup drives help protect critical systems and company data.

PART 3 — ENTERPRISE SOFTWARE INVENTORY

Software	Version	License	Purpose
Windows 11 Pro	Latest supported version	Commercial	Operating system for company desktops and laptops; provides business management and security features.
Ubuntu Server	24.04 LTS	Open Source	Server operating system for centralized services and reliable long-term support.
Microsoft Office	Microsoft 365 / current version	Subscription	Word processing, spreadsheets, presentations, email-related productivity, and business documentation.
Visual Studio Code	Latest stable version	Free / Open Source	Code editor used by IT employees for software development.
Git	Latest stable version	Open Source	Version control for tracking and managing source code changes.
GitHub Desktop	Latest stable version	Free	Graphical interface for managing Git repositories and source-code workflows.
VirtualBox	Latest stable version	Open Source	Creates virtual machines for testing systems and software.
Google Chrome	Latest stable version	Free	Web browser for web applications, research, cloud services, and daily business use.
Microsoft Defender	Built-in / current	Included with Windows	Provides malware and threat protection for Windows devices.
AnyDesk	Latest stable version	Free / Commercial options	Remote support and troubleshooting when authorized by the user and company policy.
7-Zip	Latest stable version	Open Source	Compresses and extracts files for storage and file transfer.

Software choices are aligned with the startup's software development, office productivity, system administration, security, and file-management requirements.

PART 4 — ENTERPRISE NETWORK INVENTORY

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Receives internet service from the chosen ISP.
NET-002	Router	1	Routes traffic between the office network and the internet.
NET-003	Firewall	1	Filters traffic and protects the internal network from unauthorized access.
NET-004	Managed Switch	1	Connects wired computers, server, printer, and other network devices.
NET-005	Wireless Access Point	3	Provides wireless connectivity across the office floor.
NET-006	Patch Panel	1	Organizes and terminates Ethernet network cabling.
NET-007	CAT6 Ethernet Cable	Approx. 500 m	Provides reliable wired network connections.
NET-008	RJ45 Connectors	50 pcs	Terminates CAT6 cables for Ethernet connections.

Network Design Rationale

The network uses a structured arrangement where the ISP modem connects to the router and firewall, then to the managed switch. The switch connects wired workstations, the server, printers, and access points. CAT6 cabling is selected for dependable wired connections and future expansion. The firewall provides an additional security layer between the internet and the internal company network.

PART 5 — ENTERPRISE NETWORK DIAGRAM

The final network topology will be created in Draw.io. It should use standard networking symbols, clear labels, logical connections, a diagram title, and the student's name.

Required Components

- Internet
- ISP Modem
- Router
- Firewall
- Managed Switch
- Wireless Access Point
- Server
- Printer
- IT Department
- HR Department
- Finance Department
- Sales Department

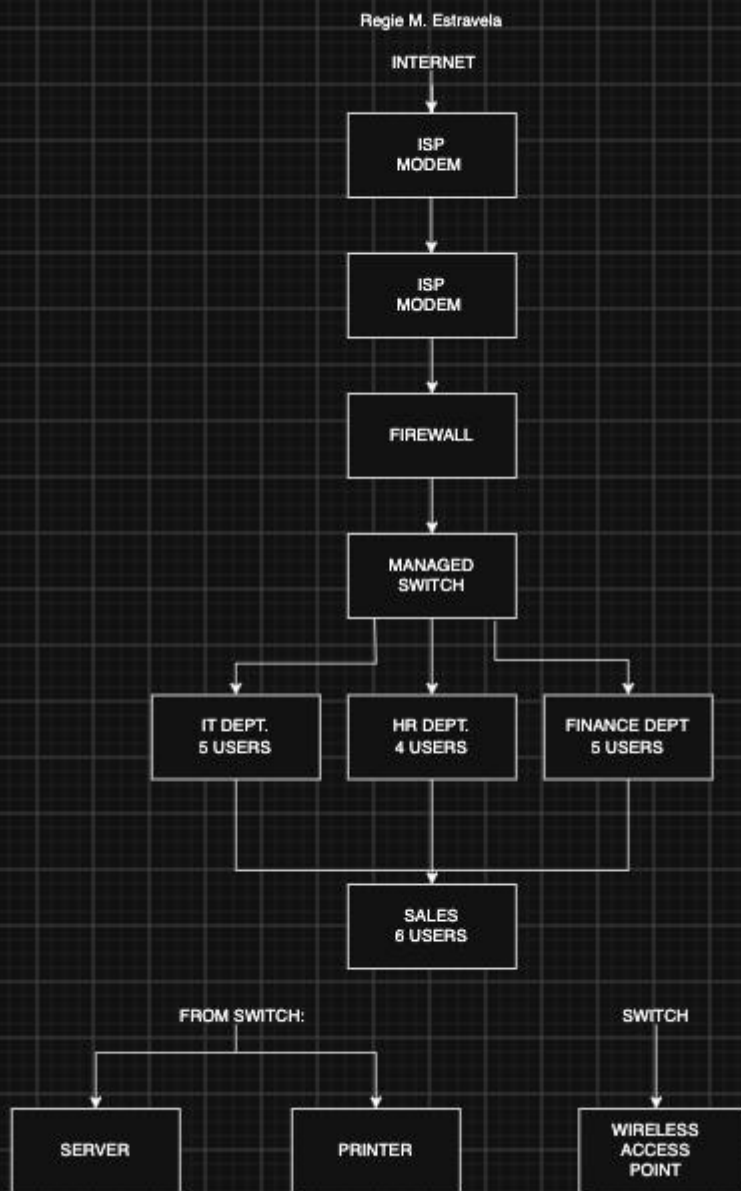
Suggested Logical Connection

Internet → ISP Modem → Router → Firewall → Managed Switch → Server / Printer / Department PCs / Wireless Access Points. The access points provide wireless connectivity while the switch provides wired connections to the departments.

Export requirements: PNG and PDF.

Note: The actual Draw.io diagram will be created after the written parts are completed.

ABC STARTUP SOLUTIONS
ENTERPRISE NETWORK TOPOLOGY



PART 6 — SYSTEM ADMINISTRATION ROLES

Helpdesk Technician

Responsibilities

Provides first-level technical support, troubleshoots hardware and software issues, assists users, documents incidents, and escalates complex problems.

Skills

Troubleshooting, communication, basic networking, operating systems, customer service, and documentation.

Common Tools

Ticketing systems, remote-support tools, Windows utilities, diagnostic tools, and knowledge bases.

Certifications

CompTIA A+ and other entry-level technical support certifications.

Network Administrator

Responsibilities

Maintains routers, switches, firewalls, wireless networks, network availability, configurations, and connectivity.

Skills

TCP/IP, routing and switching, network security, troubleshooting, monitoring, and network documentation.

Common Tools

Network monitoring tools, command-line utilities, router/switch management interfaces, packet analysis tools, and firewall consoles.

Certifications

Cisco CCNA, CompTIA Network+, and related networking certifications.

Linux System Administrator

Responsibilities

Manages Linux servers, user accounts, permissions, services, updates, storage, logs, backups, and server security.

Skills

Linux command line, shell scripting, system security, storage management, networking, and troubleshooting.

Common Tools

SSH, Bash, systemd, package managers, log utilities, monitoring tools, and virtualization platforms.

Certifications

Linux+, LPIC-1, or Red Hat-related certifications.

Cloud Administrator

Responsibilities

Manages cloud resources, user access, virtual machines, storage, networking, security controls, monitoring, and cloud backups.

Skills

Cloud computing concepts, identity and access management, networking, virtualization, security, automation, and cost awareness.

Common Tools

Cloud provider consoles, command-line interfaces, monitoring dashboards, identity management tools, and infrastructure management tools.

Certifications

AWS Certified Cloud Practitioner, Microsoft Azure Fundamentals, or comparable cloud certifications.

How the Four Professionals Work Together

The four professionals work together to keep the organization's IT environment reliable and secure. The Helpdesk Technician handles user issues and provides first-level support. When a problem involves network connectivity, the Network Administrator investigates and manages the network equipment and security controls. The Linux System Administrator maintains Linux-based servers and services that support company operations. The Cloud Administrator manages cloud resources and services when the company uses cloud infrastructure. They communicate through documentation, tickets, monitoring, and escalation so that issues are resolved efficiently and business operations remain available.

PART 7 — INFRASTRUCTURE RECOMMENDATIONS

Internet Provider

Choose a business-grade fiber internet provider available in the office area, preferably with a service-level agreement and a plan with sufficient upload and download capacity for software development, cloud services, video meetings, and daily operations. A backup internet connection should be considered for continuity.

Server Specifications

Use a business-class server with a multi-core processor, at least 32 GB ECC RAM, redundant storage using RAID, reliable network connectivity, and sufficient capacity for centralized services and future growth. The server should be placed in a secure and ventilated area and connected to a UPS.

Backup Strategy

Follow a 3-2-1 backup approach: maintain three copies of important data, on two different types of storage, with at least one copy stored offsite or offline. The NAS can provide centralized backup, while external drives can provide offline copies. Backups should be scheduled and periodically tested.

Security Recommendations

Deploy a firewall, keep systems updated, restrict administrative privileges, secure Wi-Fi, separate sensitive resources when practical, disable unused services, maintain logs, and provide security awareness training to employees.

Antivirus

Use Microsoft Defender on Windows devices and keep its security intelligence and operating system updates current. Security monitoring should be supplemented by safe browsing, patch management, access controls, and regular backups.

Password Policy

Require strong, unique passwords, prohibit password sharing, enable multi-factor authentication where available, limit administrative accounts, and change credentials when compromise is suspected. Employees should use a company-approved password manager when one is provided.

Expansion Plan

Design the infrastructure with additional switch capacity, spare cable runs, scalable storage, and room for additional wireless access points and workstations. The network and server should be reviewed regularly as the number of employees, clients, and applications grows.

PART 8 — PERSONAL REFLECTION

Working on the Week 2 Enterprise Infrastructure Planning project helped me understand that system administration is not only about fixing computers or configuring networks. Before any equipment is purchased or installed, a system administrator needs to understand the organization's requirements and plan how the different technologies will work together. Through this project, I learned how to identify hardware, software, and network requirements based on the number of employees and the functions of each department. I also learned that an infrastructure plan needs to consider security, backups, future expansion, and business continuity.

The most challenging task for me was deciding what equipment and software would be appropriate for a startup company with limited resources. It was important to avoid choosing equipment simply because it is powerful or expensive. Instead, each item needed to have a clear purpose and should support the actual work of the company. Creating the network structure was also challenging because I needed to understand the logical relationship between the internet, modem, router, firewall, switch, server, wireless access points, and departments.

This project showed me why planning is important before deployment. Proper planning can prevent unnecessary expenses, reduce compatibility problems, improve security, and make future maintenance easier. A good plan also gives management a clear idea of what the organization needs before investing in technology. Without planning, a company may purchase equipment that does not meet its requirements or create a network that becomes difficult to maintain as the company grows.

Overall, this project will help me become a better System Administrator because it developed my ability to analyze requirements, organize technical information, document infrastructure, and think about long-term solutions. It also taught me that a System Administrator should communicate technical decisions clearly and consider both the technical and business sides of an organization. I can apply these skills in future projects, internships, and actual workplace situations.